

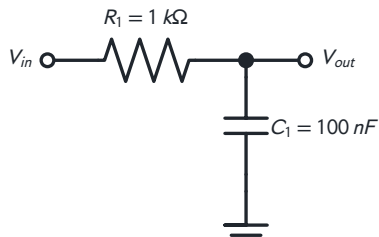
An authored netlist is resolved, solved, and returned as a canonical waveform

Direct 1 kHz RC low-pass example; output is compared with the independent steady-state transfer equation.

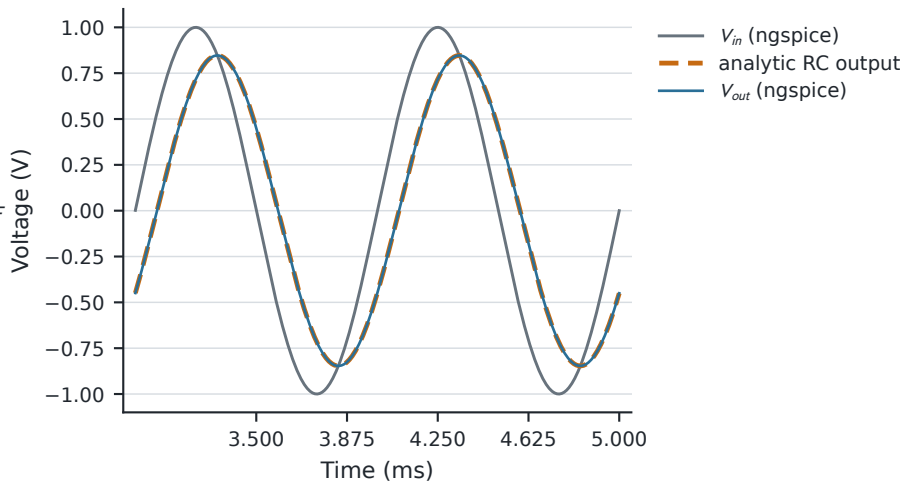
(a) Public calculation path



(b) Imported RC circuit



(c) Steady-state input and output

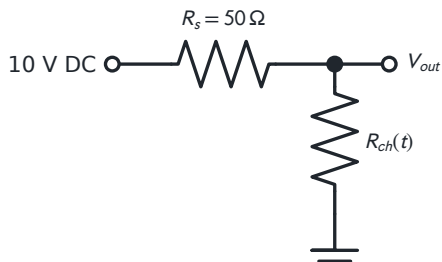


ngspice-46; last two cycles shown; analytic gain=0.846733, phase=-32.14 deg; max |error|=5.93 μV .

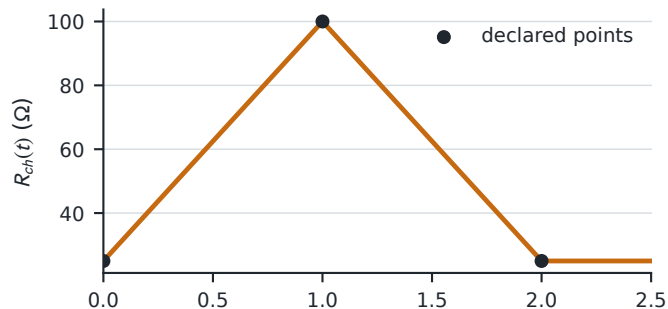
A prescribed component trajectory is used during the transient solve

The independent divider equation is evaluated at every saved time point; no fitted surrogate is used.

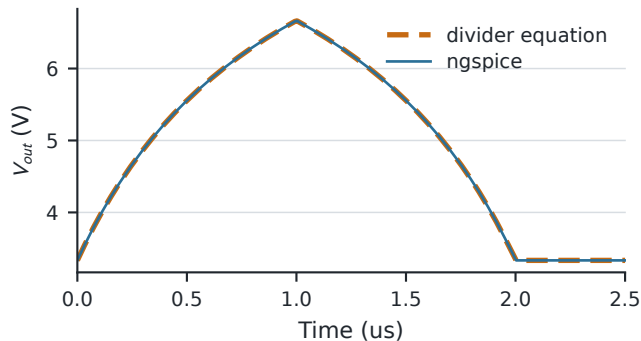
(a) Variable-resistor circuit



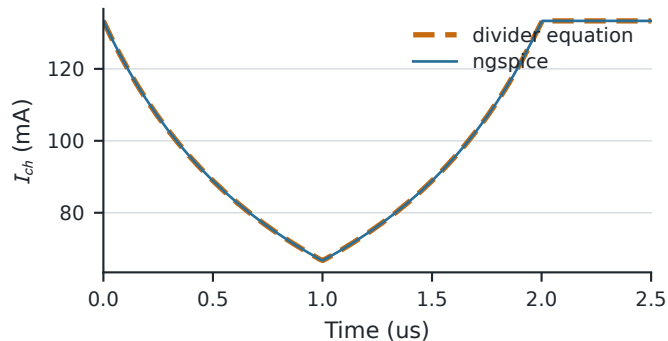
(b) Prescribed $R_{ch}(t)$



(c) Output voltage



(d) Chamber-element current



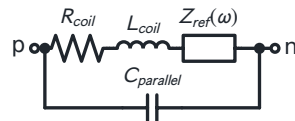
Effective plasma-load circuits propagate correctly through ngspice

Each declared load feeds the fixed benchmark L-match; points compare whole-network input Z, not plasma-state physics.

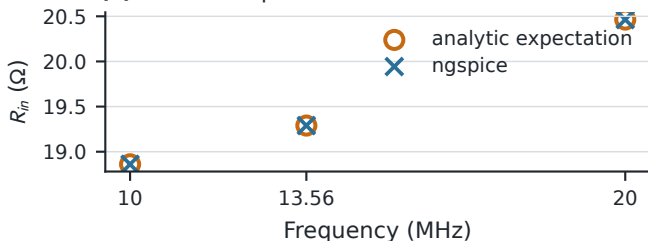
(a) Effective CCP terminal model



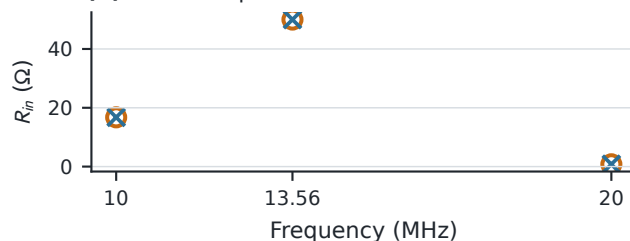
(b) Reduced ICP terminal model



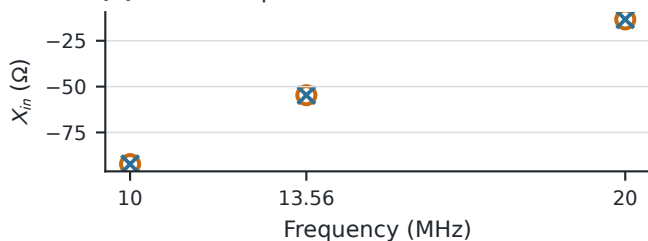
(c) CCP input resistance



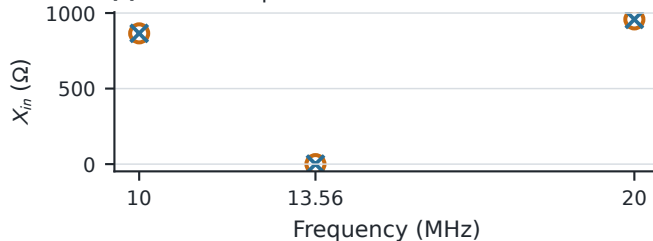
(d) ICP input resistance



(e) CCP input reactance



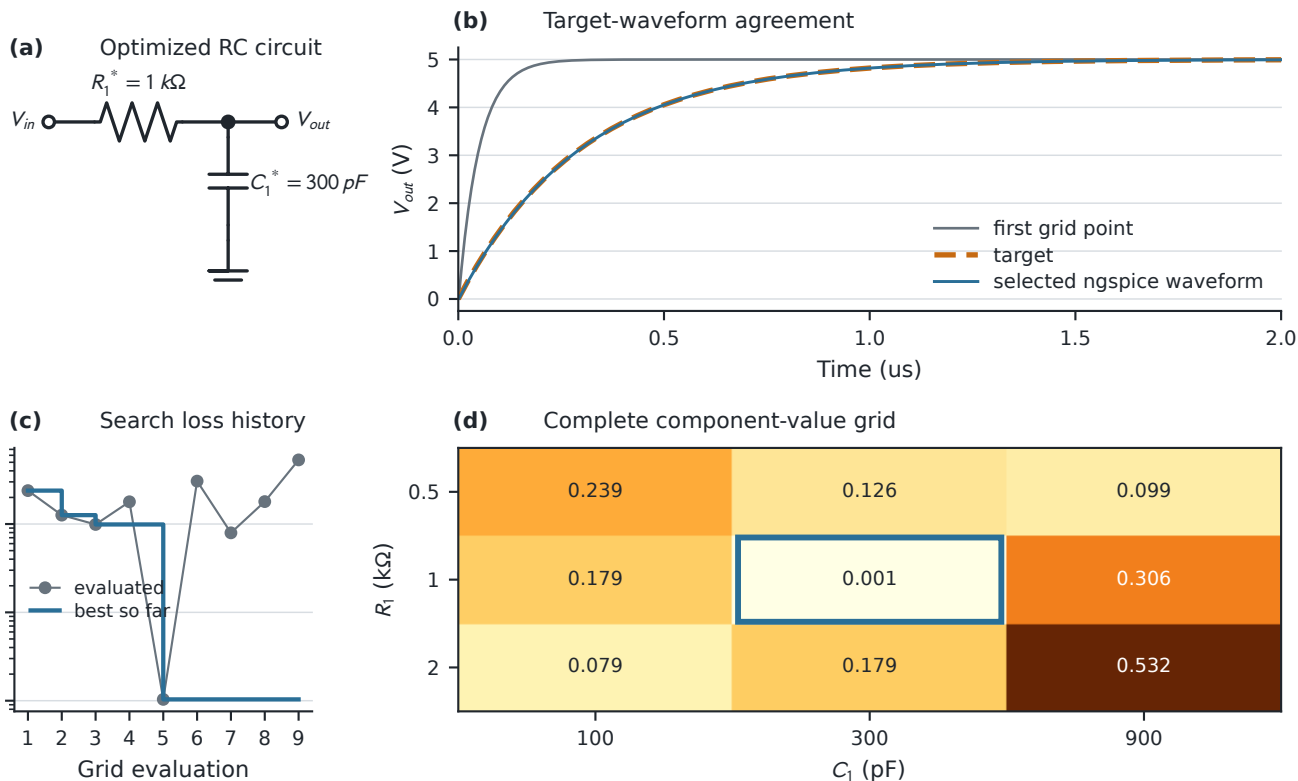
(f) ICP input reactance



Open circles: independent load + L-match equations. Crosses: ngspice extraction. Max $|Z\text{-component error}|$: CCP 4.58e-13 ohm; ICP 4.76e-10 ohm.

Element optimization recovers the RC time constant behind the target waveform

A finite 3 x 3 component grid is solved exhaustively; the selected waveform is an ngspice result, not a surrogate prediction.



Selected $R_1=1 \text{ k}\Omega$, $C_1=300 \text{ pF}$, $\tau=300 \text{ ns}$; nRMSE=0.0010; 9/9 ngspice evaluations succeeded.